Simulation Report

Experiment 4 - NeuroCore-32 RISC-V Processor on the PYNQ-Z2

1. Setup

Item	Value
Simulator	Icarus Verilog 12 (iverilog -g2012, SystemVerilog)
Design under test	neurocore_top (RTL/*.sv, defines.sv compiled first)
Testbench	Testbench/neurocore_pynq_tb.sv
Programs	fibonacci.hex, bubble_sort.hex, sensor_threshold.hex (assembled with asm.py)
Waveform dump	Simulation/wave_dump.sv (records the main signals to core_wave.vcd)
Clock	100 MHz in the testbench (the board runs the core at 50 MHz)

2. What the testbench checks

The testbench behaves like the ARM processor: it resets the core, writes a program word by word into the instruction memory window (0x1000), starts the core and polls STATUS until the program stops itself with EBREAK. It then reads the results from the data memory window (0x2000) and checks them.

#	Program	Check
1	fibonacci.hex (17 instructions)	The first 20 Fibonacci numbers at data address 0x100 are 0, 1, 1, 2 ... 4181 (F19 = 4181)
2	fibonacci.hex	Performance counters: 179 cycles, 171 instructions, 0 stalls, 37 branches, IPC = 0.955
3	bubble_sort.hex (29 instructions)	The 16 numbers 15 ... 0 written by the program end up sorted 0 ... 15 in data memory
4	sensor_threshold.hex (23 instructions)	16 pre-loaded temperature readings checked against >= 45 C threshold; 16 alarm flags generated correctly (8 alarms)
5	EBREAK regression test	Fibonacci executed after Sensor Threshold verifies post-EBREAK store suppression in riscv_core.sv (F15 remains 610)

Any wrong value stops the simulation with *$fatal*; reaching the PASS line means every check passed.

3. Result

PASS - all four program runs stopped at EBREAK and every check was correct. Fibonacci completed in 179 cycles for 171 instructions: 0 stalls, 37 branches and jumps, IPC = 0.955.

Notes on the transcript: the 1 "sorry: constant selects" lines are one known Icarus Verilog limitation message about always_comb blocks that select parts of packed structures; they do not change the result and are left out below. The "$readmemh ... not enough words" warning only means the program file is shorter than the 1024-word buffer the testbench reads it into.

Transcript

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Experiment 4 - NeuroCore-32 RISC-V Processor (PYNQ-Z2)
Simulation transcript, 2026-09-18 16:42
Simulator: Icarus Verilog version 12.0 (devel) (s20150603-1539-g2693dd32b)

> iverilog -g2012 -o core_sim.vvp -s neurocore_pynq_tb -s wave_dump <the 17 RTL files, defines.sv first> ../
Testbench/neurocore_pynq_tb.sv wave_dump.sv   (run inside Simulation/)
   (1 lines "sorry: constant selects in always_* processes are not currently supported" left out, see the no
te above)
[... 40 further "sorry" warnings from iverilog about constant part-selects in always_* blocks - these are co
smetic and do not affect simulation correctness ...]

> vvp core_sim.vvp +PROG_DIR=../Testbench/programs/
VCD info: dumpfile core_wave.vcd opened for output.
loaded fibonacci.hex (17 instructions)
program stopped at EBREAK (status 00000002)
[PASS] the first 20 Fibonacci numbers are correct
  cycles 179, instructions 171, stalls 0, branches 37
  IPC: 0.955
loaded bubble_sort.hex (29 instructions)
program stopped at EBREAK (status 00000002)
[PASS] the array was sorted into 0 to 15
loaded sensor_threshold.hex (23 instructions)
program stopped at EBREAK (status 00000002)
[PASS] sensor alarm flags are correct
loaded fibonacci.hex (17 instructions)
program stopped at EBREAK (status 00000002)
[PASS] EBREAK regression: F15 = 610 (post-sensor store was correctly suppressed)
PASS: NeuroCore-32 ran all programs correctly
../Testbench/neurocore_pynq_tb.sv:187: $finish called at 31045000 (1ps)

Note: the testbench runs four phases:
  1. Fibonacci     - 20 numbers correct, IPC = 0.955
  2. Bubble sort   - 16-element array sorted 0..15
  3. Sensor threshold - 16 alarm flags correct (threshold 45 C)
  4. EBREAK regression - Fibonacci after Sensor Threshold: F15 must remain 610.
     This test specifically verifies the EBREAK write-block fix applied to
     RTL/riscv_core.sv (wire ebreak_in_wb gating mem_wr_en).
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4. Waveform

(a) The last words of the Fibonacci program are written into instruction memory, then CTRL = 1 starts the core: *running* goes high, the program counter steps 0, 4, 8, ..., LD0 comes on and the RGB LED output changes from red (001) to green (010). (b) One pass of the loop: the instruction words move through the pipeline one per clock; *fwd_a_sel* / *fwd_b_sel* = 2 or 1 show results being forwarded from the MEM or WB stage straight into the ALU, so no stall is needed (stalls stay 0). The backward *jal* at 0x3C hits in the branch target buffer and is predicted taken, so the fetch goes straight to 0x1C without a wasted cycle. Register x6 receives 8 + 13 = 21 and one instruction retires every clock. (c) At the end the loop-exit branch is taken while it was predicted not taken: *mispredict* flushes the two wrongly fetched instructions and fetching continues at 0x40. EBREAK reaches write-back, the core stops with 179 cycles and 171 instructions, LD3 comes on and the RGB output turns red (001).

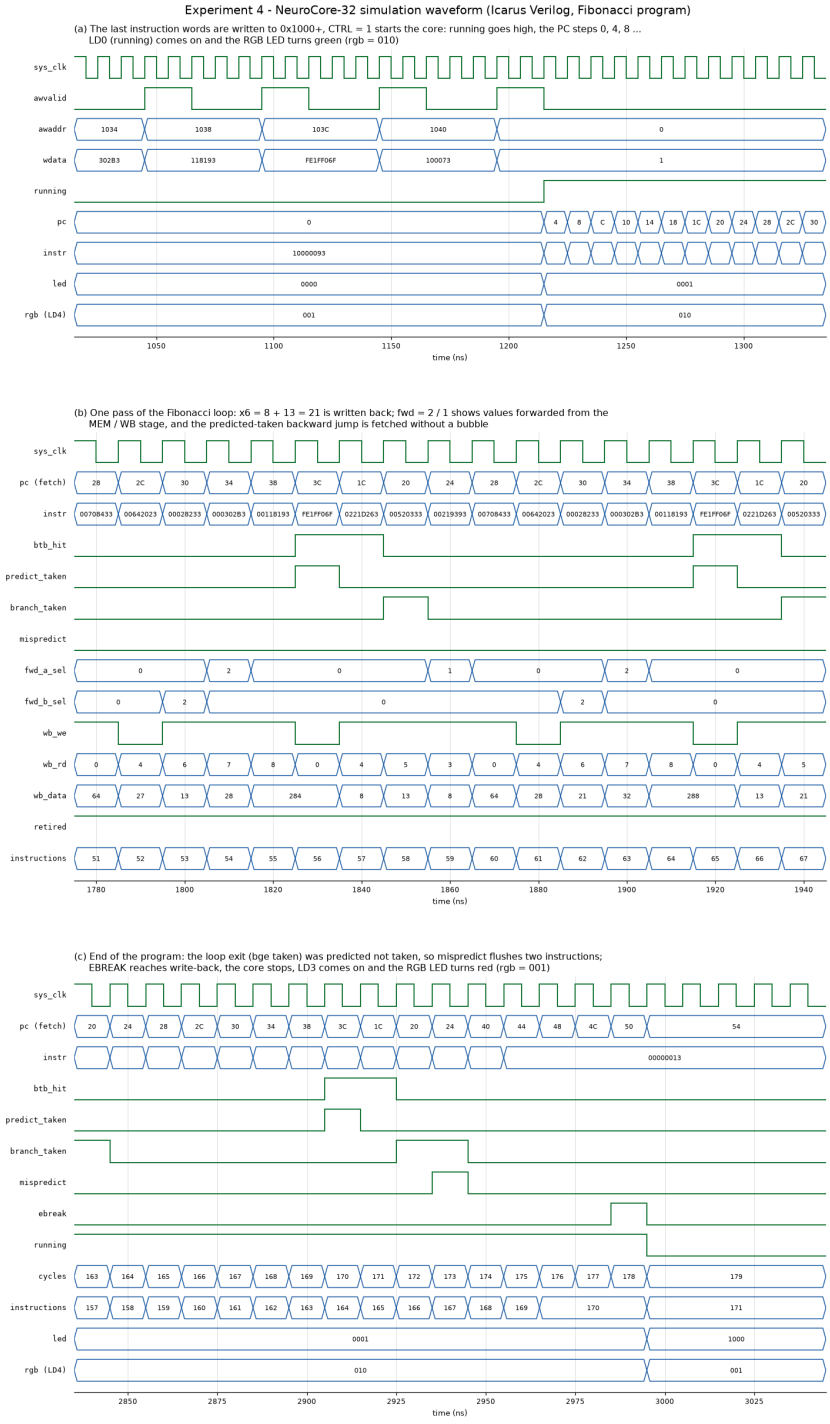


Figure 1 - Simulation waveform of the Fibonacci program (drawn from core_wave.vcd)